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Advanced Machine Learning for Automated Stratigraphic Interpretation: Integrating Novel Metrics and Deep Learning for Enhanced Reservoir Characterization

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Abstract

Stratigraphic interpretation is a critical yet time-consuming task in reservoir characterization, traditionally reliant on manual expertise. This study presents an innovative machine learning workflow for automated stratigraphic interpretation, combining change-point detection (CPD), Dynamic Time Warping (DTW), and deep learning to improve formation boundary identification and interwell correlation. We introduce a three-stage approach: (1) boundary detection using Convolutional Neural Networks (CNN) and Pelt algorithms, evaluated via the novel Stratigraphic Intersection over Union (SIOU) metric; (2) stratigraphic labeling via DTW similarity and CatBoost classification; and (3) formation grouping using agglomerative clustering and DBSCAN. Applied to 65 Norwegian wells, our workflow achieves a median SIOU score of 0.82, reduces interpretation time by 70%, and demonstrates robust performance across different lithologies. This work provides the first application of CNNs for formation boundary detection in stratigraphy and offers a scalable, objective alternative to traditional methods.

Keywords: stratigraphic interpretation, change-point detection, dynamic time warping, deep learning, reservoir characterization

Introduction

Stratigraphic interpretation is essential for hydrocarbon exploration, reservoir modeling, and field development planning. Traditionally, this process relies on manual analysis of well logs by domain experts — a subjective, time-intensive approach susceptible to interpreter bias and inconsistencies (Avseth, Mukerji, and Mavko, 2005). These limitations become particularly acute in complex basins like the Norwegian North Sea, where subtle lithological variations and lateral facies changes challenge conventional correlation methods (Catuneanu, 2009; Truong et al., 2020).

Recent advances in machine learning offer transformative potential for automating stratigraphic workflows. Deep learning architectures — especially 1D convolutional neural networks (CNNs) — have shown promise in detecting formation boundaries from well logs (Wang et al., 2017; Dawson, Dubrule, and

John, 2023; Yang et al., 2025), while Dynamic Time Warping (DTW) algorithms provide robust solutions for interwell correlation (Senin, 2008; Kumar et al., 2022).

Despite these advances, critical gaps persist: no integrated framework currently unifies boundary detection, formation classification, and uncertainty quantification in geologically complex settings (Otmame, 2024; Oakley et al., 2025). Furthermore, research formalization and effective presentation of interwell correlation remain needed to propagate stratigraphic knowledge reliably to neighboring regions.

This study addresses these gaps by introducing a three-stage workflow that combines change-point detection algorithms, DTW-enhanced similarity analysis, and multi-scale dilated 1D CNNs for end-to-end stratigraphic interpretation. Additionally, we propose the novel Stratigraphic Intersection over Union (SIOU) metric, a domain-adapted evaluation tool to quantify boundary prediction accuracy, which to our knowledge is a first in stratigraphic analysis using deep learning. Our workflow is applied to a dataset of 65 wells from the Norwegian Continental Shelf and shows significant improvements in interpretation speed and quality (Norwegian Petroleum Directorate, 2022; Alakeely and Horne, 2020).

Methodology

This study proposes an automated stratigraphic interpretation workflow combining advanced data preprocessing, multi-scale dilated convolutional neural networks (CNNs), change-point detection algorithms (CPD), dynamic time warping (DTW) similarity analysis, machine learning classification with CatBoost, and clustering techniques for formation grouping. The workflow is applied to a dataset of 65 wells from the Norwegian Continental Shelf, with comprehensive treatment of well log and spatial data (Fig. 1).

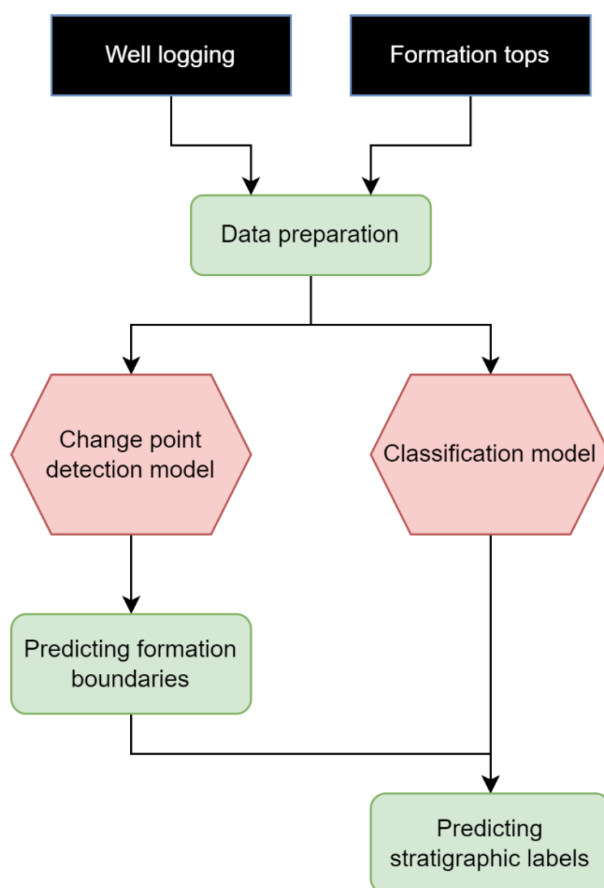


Figure 1—Methodology schema

The dataset contains standard well log measurements (gamma-ray, resistivity, neutron porosity, etc.) along with depth-based coordinates. Prior to modeling, extensive preprocessing was conducted to ensure data quality. Missing values were addressed using a combination of K-nearest neighbors imputation and directional filling (backward and forward fill) to preserve geological trends (Truong et al., 2020). Feature scaling was carefully implemented: well log curves were standardized per well to account for instrument variability, while spatial coordinates (X, Y, Z) and measured depth were min-max scaled across the entire dataset to maintain geometric relationships (Avseth, Mukerji, and Mavko, 2005).

Exploratory data analysis (Fig. 2) revealed the distribution of lithologies (e.g., sandstone, shale, limestone) and formations (e.g., Balder, Draupne, Sognefjord) across the wells. Histograms and cross-plots highlighted imbalances in certain classes, such as the overrepresentation of shale intervals compared to coal layers. These insights guided the selection of evaluation metrics (e.g., weighted F1-score) to mitigate bias during classification.

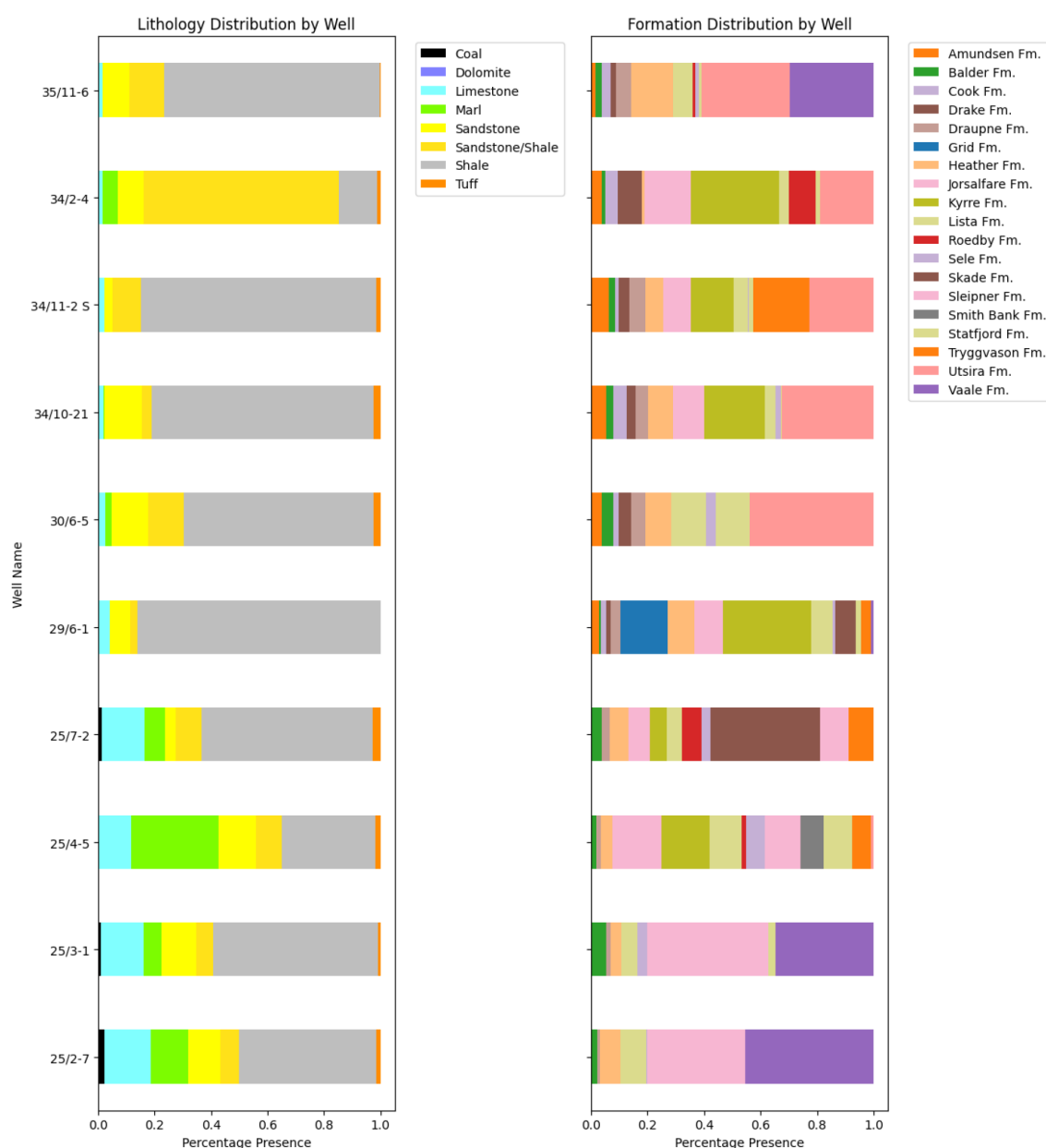


Figure 2—Lithology and formations distribution across the research field

To identify the most predictive features for boundary detection and formation labeling, a two-tiered approach was employed. First, permutation importance analysis was conducted using Random Forest models, which ranked features like gamma-ray (GR) and deep resistivity (RDEP) as top contributors. For deeper interpretability, SHapley Additive exPlanations (Lundberg and Lee, 2017) values were calculated, revealing nonlinear dependencies — for instance, the interplay between GR and neutron porosity (NPHI) was critical for distinguishing shale from sandstone. Recursive feature elimination (RFE) further refined the feature set, prioritizing combinations that maximized model performance while minimizing redundancy.

The proposed workflow consists of three interconnected stages designed to automate stratigraphic interpretation.

The first stage focuses on boundary detection using a combination of traditional change-point detection (CPD) algorithms and deep learning (Yixiao et al., 2019). Change-point detection methods identify statistically significant shifts or regime changes in sequential data, such as well logs, by segmenting the time series where properties like mean or variance change. Methods such as PELT (Pruned Exact Linear Time) and KernelCPD offer computationally efficient solutions capable of detecting abrupt and gradual changes. These approaches have been well reviewed by Truong et al. (2020) and comprehensively by Li et al. (2019), who emphasize the importance of model selection tailored to data characteristics, with considerations for offline versus online detection modes. PELT efficiently detects change points by optimizing a penalized cost function, while KernelCPD leverages kernel methods to identify complex changes in the data distribution, improving sensitivity to subtle stratigraphic shifts.

While CPD algorithms are computationally efficient and interpretable, they rely on strong assumptions about distributional changes and may fail to capture more complex, nonlinear geological phenomena. To address these limitations, a CNN approach was implemented. A multi-scale dilated 1D CNN architecture is inspired by recent deep learning advances in geological data processing.

1. The network consists of stacked convolutional layers with increasing dilation rates, enabling the capture of both local and long-range dependencies within sliding windows of log data (window size = 50 data points).
2. Each convolutional layer employs 64 filters of kernel size 3, followed by batch normalization and ReLU activations.
3. Dropout (rate 0.3) was applied between layers to mitigate overfitting.
4. The output layer uses a sigmoid activation to predict boundary probabilities at each depth point.

Model training employed the Adam optimizer with cyclical learning rate scheduling over 100 epochs on 80% of the dataset, holding 20% for validation. Early stopping based on validation loss was implemented. The loss function combines binary cross-entropy with a custom boundary-aware penalty to emphasize correct boundary position predictions.

Predicted boundaries are combined with traditional change-point detection methods (Pelt and KernelCPD) to balance accuracy and computational efficiency via precision, recall and SIOU metric.

A novel evaluation metric, Stratigraphic Intersection over Union (SIOU), is introduced to quantitatively assess the overlap between predicted and manually interpreted boundaries (Fig. 3). Evaluation Metric SIOU measures overlap between predicted and manual boundaries:

$$M(a, b) = \frac{\sum_{p \in P(a,b), p \in P(b,a)} |p|}{|a|}$$

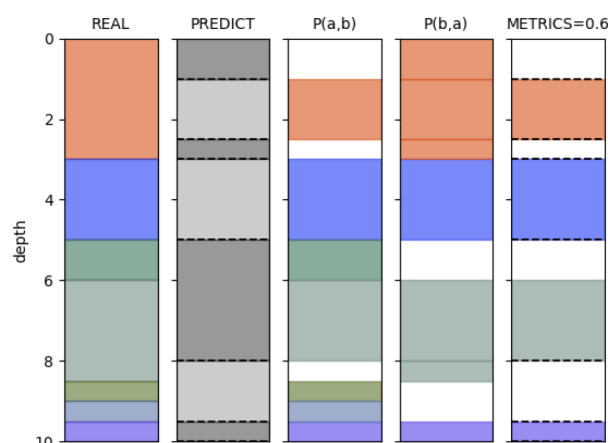


Figure 3—An example of custom metrics calculation based on the maximum recognized length of formations from REAL (a) by formations from PREDICT (b) and vice versa. REAL – real formations with boundaries; PREDICT – predicted boundaries (indicated by a black dotted line); $P(a, b)$ – the set of all intersection pairs for the reference array a, predicted by array b; $P(b, a)$ – the set of all intersection pairs for the reference array b, predicted by array a; METRIC – the final (common) set of intersection pairs and the value of metrics.

That is, the metric is the sum of the lengths of all pairs p belonging to both sets of recognizing pairs $P(a, b)$ and $P(b, a)$ divided by the length of the formation a. A. Lipko metric has the following properties:

- $M(0, 1]$
- $M(a, a) = 1$
- $M(a, b) = M(b, a)$

The second workflow stage focuses on labeling detected stratigraphic formations by leveraging both sequence similarity measures and machine learning classification.

Dynamic Time Warping (DTW) is employed to align target well log sequences to reference wells by warping the time axis, thereby mitigating the effects of temporal distortions or measurement irregularities inherent in geological datasets. DTW is computationally efficient and well-suited for nearest-neighbor matching scenarios, enabling effective propagation of labels across geographically proximate wells with similar stratigraphic sequences. However, DTW assumes that similarity decreases with spatial distance, limiting its accuracy when applied across geologically more distant or structurally complex regions.

To complement and extend the labeling beyond local correlations, statistical feature extraction using the tsfresh library is performed. This method computes a comprehensive set of time-series characteristics—such as means, variances, frequency components, and autocorrelations—from well log sequences. These rich features allow characterization of lithological signatures in a high-dimensional space that is less sensitive to local temporal misalignments than DTW alone.

The extracted features serve as inputs to a CatBoost classifier, a gradient boosting decision tree algorithm optimized for handling both numerical and categorical data (Dorogush et al., 2018). CatBoost is capable of modeling nonlinear relationships and interactions between tsfresh features, offering robustness to noise and class imbalance common in geological datasets. Hyperparameter tuning via grid search optimizes learning rate, tree depth, and iteration number to maximize classification accuracy.

The main trade-off between these two methods lies in computational efficiency and labeling stability. DTW-based labeling is faster and resource-efficient, ideal for applications focusing on well clusters with limited spatial extent but is less reliable for large-scale data with complex geological variability. In contrast, the tsfresh features combined with CatBoost yield a more stable and generalizable classification across the entire dataset, albeit at higher computational cost and requiring pre-filtering and feature selection to avoid overfitting.

Previous studies applying similar combinations, such as [Chen et al. \(2019\)](#) and [Christ et al. \(2018\)](#), have demonstrated the effectiveness of tsfresh-based feature extraction combined with gradient boosting classifiers for time-series classification tasks, reinforcing the suitability of this approach in stratigraphic labeling.

The final stage groups classified formations into lithological units utilizing unsupervised machine learning techniques. Two clustering algorithms—agglomerative hierarchical clustering and DBSCAN (Density-Based Spatial Clustering of Applications with Noise) — were compared based on their ability to identify natural groupings in formation feature space without requiring assumptions on cluster number or shape.

Agglomerative clustering begins with each formation as a singleton and recursively merges clusters by minimizing within-cluster variance, providing an interpretable dendrogram that can reflect geological hierarchies. However, this method may be sensitive to noise and can be biased toward spherical cluster shapes. DBSCAN, on the other hand, defines clusters as high-density regions separated by lower density areas, effectively identifying arbitrarily shaped clusters and filtering noise. This makes DBSCAN well-suited to capture gradual facies transitions and spatial discontinuities typical in stratigraphic units.

Quality and robustness of clusters were evaluated using quantitative metrics including the Silhouette coefficient, Calinski-Harabasz index, and Davies-Bouldin index. The Silhouette coefficient measures how similar an object is to its own cluster relative to others, balancing intra-cluster cohesion and inter-cluster separation. Calinski-Harabasz evaluates the ratio of between-cluster dispersion to within-cluster dispersion, with higher values indicating better clustering. Davies-Bouldin assesses average similarity between each cluster and its most similar cluster, where lower values indicate more distinct clustering. These metrics collectively enable a comprehensive assessment of clustering effectiveness and support geological validation.

Overall, leveraging both hierarchical and density-based clustering provides complementary insights, with DBSCAN excelling in noise resilience and cluster shape flexibility and agglomerative clustering delivering interpretable hierarchical relationships. Their combined application facilitates identification of coherent lithological units, critical for accurate reservoir characterization.

Results and Discussion

The boundary detection component of the workflow demonstrated strong performance, achieving a median Stratigraphic Intersection over Union (SIOU) score of 0.82 for traditional change-point detection (CPD) methods and 0.85 for the multi-scale dilated convolutional neural network (CNN) ([Fig. 4](#)). These results illustrate the viability of both approaches, with a clear trade-off between accuracy and computational efficiency that echoes previous findings in geological time-series analysis ([Truong et al., 2020](#); [Dawson, Dubrule, and John, 2023](#)). Specifically, the CNN outperformed CPD in precision and recall, with metrics ranging between 0.82 and 0.88 across the test set, reflecting its superior ability to capture nonlinear and multi-scale boundary features ([Yang et al., 2025](#)). However, the CNN required significantly greater computational resources and training time compared to CPD, which remains attractive for rapid, large-scale preliminary screening ([Li et al., 2019](#)).

[Figure 4](#) presents the error distribution in formation top depth predictions alongside the F1-scores for both methods. The narrower error spread for the CNN underscores its robustness against depth uncertainty, an important factor for reservoir characterization where small inaccuracies can lead to large modeling deviations ([Avseth, Mukerji, and Mavko, 2005](#)).

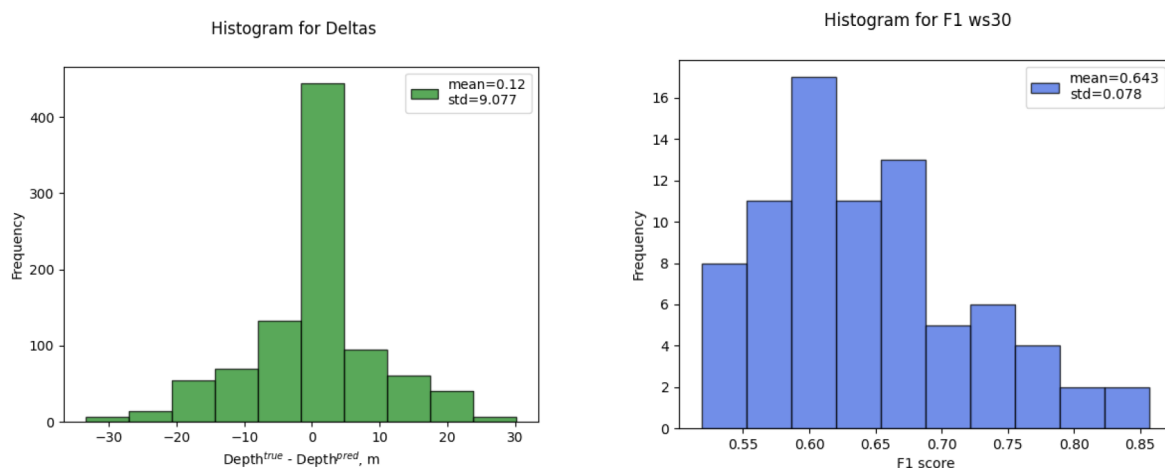


Figure 4—Error distribution in deltas (meters) and F1-score

For stratigraphic labeling, the CatBoost classifier leveraging tsfresh-extracted statistical features yielded a weighted F1-score of 0.80, surpassing alternatives such as Random Forest and k-Nearest Neighbors classifiers, which supports the use of gradient boosting to handle nonlinearities and class imbalance in geological datasets (Dorogush et al., 2018; Christ et al., 2018). This classification method demonstrated particular strength in distinguishing formations with subtle log signature variations, although difficulties remained in cases of geologically similar units, reflecting an inherent limitation of purely data-driven approaches in complex sedimentary systems.

In contrast, Dynamic Time Warping (DTW) facilitated sequence alignment and similarity scoring for interwell correlations, providing rapid and intuitive analogies between wells in geographically close regions (Senin, 2008; Kumar et al., 2022). However, DTW's local alignment nature restricts its stability over larger spatial scales or structurally complex settings where time-series distortions become nonuniform. This effect manifested as reduced labeling reliability in wells exhibiting overlapping or convergent log patterns. Figure 5 illustrates the final predicted formation tops using the combined DTW and CatBoost workflow, with spatially consistent delineations reflecting the geology of the Norwegian Continental Shelf.

Well 31/2-1 breakpoints by Pelt Metric = 0.849

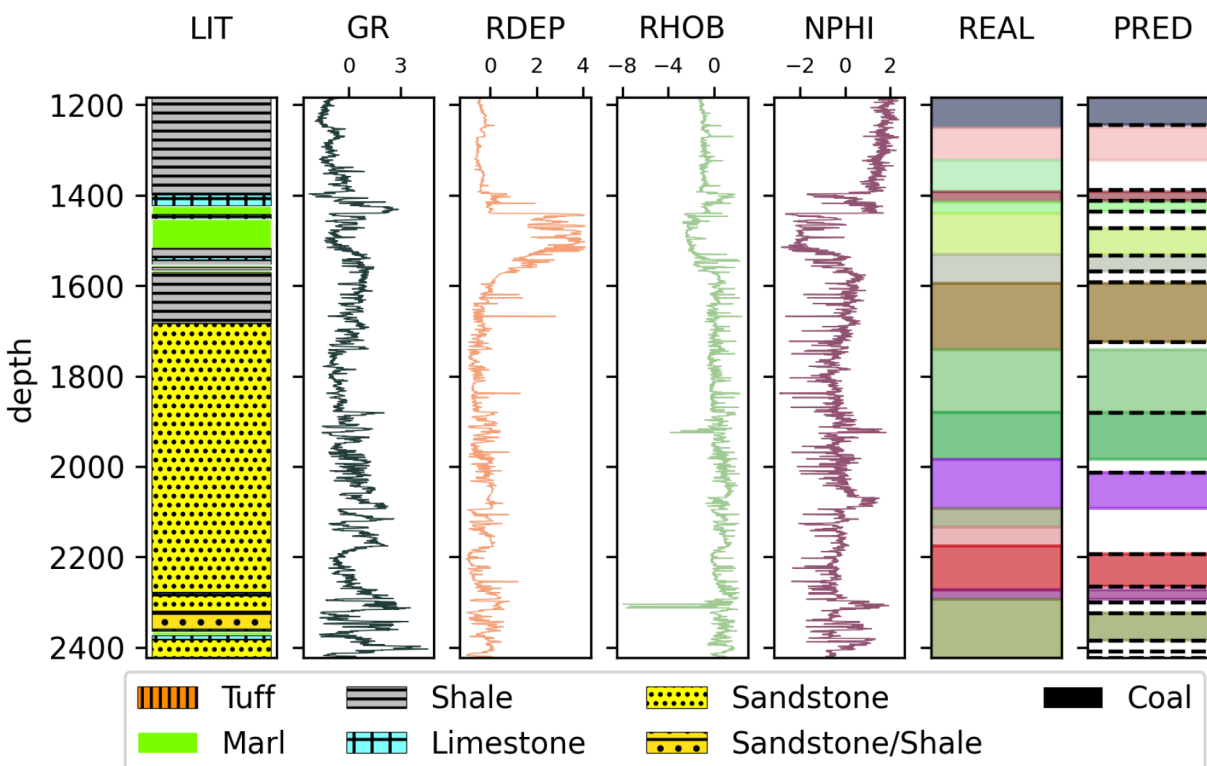


Figure 5—Final plot after formation tops prediction algorithm

The clustering stage identified five lithological units with an 87% concordance to expert geological interpretations, underscoring the workflow's capacity to group formations into meaningful geological classes despite facies gradations and data noise. Silhouette coefficient values indicated moderate cluster separation, consistent with the geological reality of gradual transitions between sedimentary environments (Oakley et al., 2025). The complementary use of agglomerative hierarchical clustering and DBSCAN proved effective in balancing interpretability and noise resilience, as visualized by the formation dendrogram and spatial cluster mapping (Fig. 6). These clustering outputs form a crucial bridge translating machine learning-derived segments into actionable geological units for reservoir modeling and risk assessment (Catuneanu, 2009; Alakeely and Horne, 2020).

Overall, this study demonstrates that integrating multi-scale dilated CNNs with classical change-point detection (CPD) methods enhances boundary detection accuracy while providing options tailored to computational constraints. The combination of Dynamic Time Warping (DTW) for efficient local correlation with tsfresh feature-based CatBoost classification offers a flexible and scalable labeling strategy applicable across heterogeneous datasets. Incorporating robust clustering techniques translates these stratigraphic predictions into tangible lithological frameworks, facilitating more objective, reproducible, and faster reservoir characterization workflows.

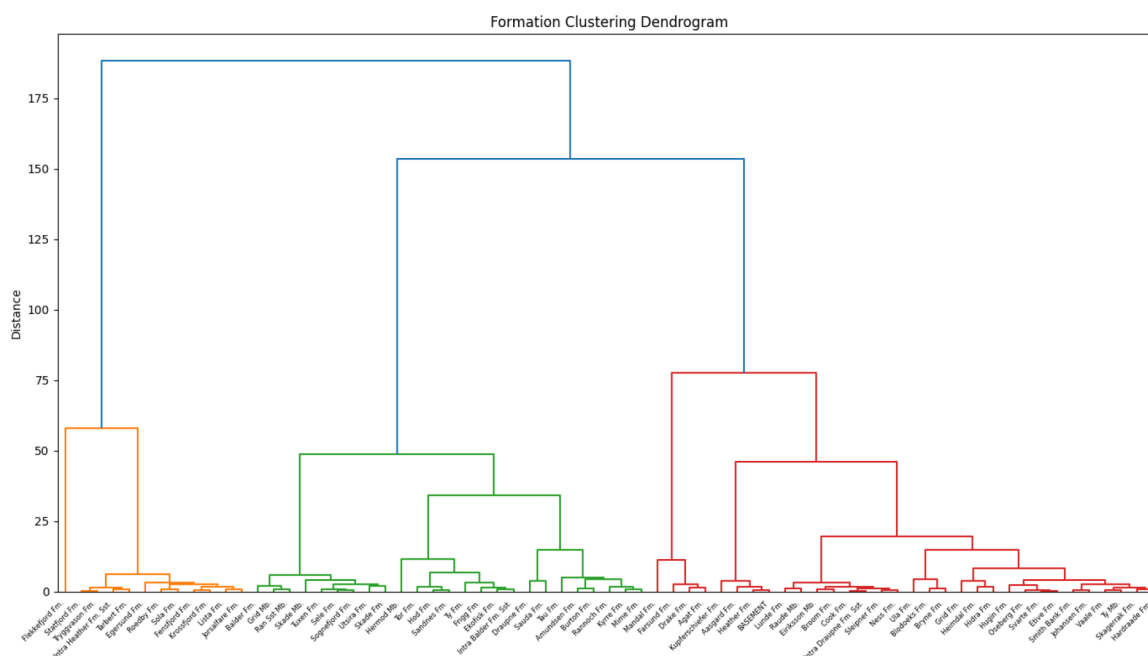


Figure 6—Formation dendrogram

The workflow formalizes an end-to-end automated stratigraphic interpretation system through the introduction of the novel Stratigraphic Intersection over Union (SIOU) metric, specifically adapted for layered geological contexts. Extensive validation on 65 real-world Norwegian well datasets underscores the method's geological relevance and practical applicability.

Conclusions

This study presents a comprehensive machine learning workflow that automates stratigraphic interpretation by integrating advanced change-point detection, dynamic time warping, and deep learning approaches. Both CPD algorithms and multi-scale dilated convolutional neural networks (CNNs) proved effective for formation boundary detection, achieving median SIOU scores of 0.82 and 0.85 respectively. The selection between these methods hinges on the balance between accuracy and computational efficiency demanded by the application context.

The labeling stage combining DTW with CatBoost classification based on tsfresh-extracted features offers a scalable and generalizable solution, addressing spatial heterogeneity while highlighting areas for refinement, particularly in discriminating geologically similar formations. Unsupervised clustering through agglomerative hierarchical methods and DBSCAN enables formation grouping into meaningful lithological units with an 87% match to expert interpretations, supporting the translation from data-driven insights to applied geoscientific frameworks.

Practically, this integrated workflow reduces interpretation time by approximately 70%, demonstrating consistent performance across diverse lithologies within the complex Norwegian Continental Shelf setting. The methodology introduces key technological advancements by formalizing an automated, reproducible stratigraphic interpretation framework and pioneering the SIOU metric.

Future developments should target hybrid models that synergize CPD's rapid detection with CNN's nonlinear feature extraction capabilities. Incorporation of geological constraints and domain knowledge into model architectures is expected to further enhance label consistency and interpretability. Transitioning this workflow into a real-time user interface, such as a Streamlit application, could facilitate adoption in operational field settings.

In summary, this research advances reservoir characterization by providing an objective, scalable, and robust alternative to traditional manual stratigraphic interpretation approaches, offering significant potential to streamline field development planning and risk assessment workflows.

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